

Term Paper: Advanced Condensed Matter BCS–BEC Crossover

Soudip Kundu

November 2025

Contents

1	Introduction	2
2	Bose-Einstein Condensate	2
2.1	Ideal Bose Gas	3
3	Cooper Pair	4
4	Ground state of the interacting fermi gas	5
4.1	Low Energy Excitation	9
4.2	Pair size	11
5	Fermionic Pair Formation and Condensation	13
5.1	Pair Formation Temperature	13
5.2	Critical Temperature for Condensation	14
6	Pair Formation and Condensation through functional integration formulation	15
6.1	Saddle Point Approximation	16
6.2	Beyond Saddle point	17
7	Appendix	17
7.1	Relation of the scattering length with scattering amplitude . . .	17
7.2	Relation between U (scattering potential) and a_s (scattering length)	19

1 Introduction

Bose-Einstein Condensation (BEC) is a well-known phenomenon for bosons, understood from the ideal Bose gas. In contrast, condensation in an ideal Fermi gas is not possible due to the Pauli principle, which forbids the ground state to be occupied by more than one fermions. However, BEC is found to occur in an interacting Fermi gas. The BEC and BCS phases of an interacting Fermi gas can be understood as two opposite limits in the description of the gas. A smooth crossover from one phase to the other is possible by tuning the scattering length/interaction strength of the gas, where the BCS limit is given by an infinite and negative scattering length and the BEC limit by an infinite and positive one. The transition from BCS to BEC is rather a crossover than a phase transition, because it is a smooth transition i.e, there is no abrupt change in the order of the physical parameter when going from one limit to the other.

2 Bose-Einstein Condensate

Before we dive into the transition let us first understand what Bose-Einstein Condensate is? If we consider a Bose gas in grand canonical ensemble, then the partition function for each state $|r\rangle$ is given as

$$Z = \sum_{\{n_r\}} e^{-\beta n_r (E_r - \mu)} = \frac{1}{1 - e^{-\beta(E_r - \mu)}} \quad (1)$$

Notice that we've implicitly assumed that the sum above converges, which is true only if $(E_r - \mu) > 0$. But this should be true for all states E_r . We will set the ground state to have energy $E_0 = 0$, so the grand partition function for a Bose gas only makes sense if

$$\mu < 0 \quad (2)$$

Now we use the fact that the occupation of one state is independent of any other. The full grand partition function is then

$$Z = \prod_r \frac{1}{1 - e^{-\beta(E_r - \mu)}}. \quad (3)$$

From this we can compute the average number of particles,

$$N = \frac{\partial}{\partial \mu} (\ln Z) = \sum_r \frac{1}{e^{\beta(E_r - \mu)} - 1} \equiv \sum_r \langle n_r \rangle. \quad (4)$$

Here $\langle n_r \rangle$ denotes the average number of particles in the state $|r\rangle$,

$$\langle n_r \rangle = \frac{1}{e^{\beta(E_r - \mu)} - 1}. \quad (5)$$

This is the Bose-Einstein distribution. Here we use

$$z = e^{\beta \mu} \quad (6)$$

since $\mu < 0$ we have $0 < z < 1$. Just keep this at the back of your mind we will need this

2.1 Ideal Bose Gas

Let us look again at a gas of non-relativistic particles, now through the eyes of quantum mechanics. The energy is

$$E = \frac{\hbar^2 k^2}{2m}. \quad (7)$$

The density of states is given as

$$g(E) = \frac{V}{4\pi^2} \left(\frac{2m}{\hbar^2} \right)^{3/2} E^{1/2}. \quad (8)$$

From this, together with the Bose-Einstein distribution, we can easily compute the total number of particles in the gas,

$$N = \int_0^\infty dE \frac{g(E)}{z^{-1}e^{\beta E} - 1}, \quad (9)$$

where $z = e^{\beta\mu}$ is the fugacity.

Next, we ask what happens at low temperature ? Recall that for convergence of the grand partition function, we require $\mu < 0$. Since the high temperature limit is $z \rightarrow 0$, we'll anticipate that quantum effect and low temperatures come about as $z \rightarrow 1$

Recall that the number density is given by (9), which we write as

$$\frac{N}{V} = \frac{1}{4\pi^2} \left(\frac{2mk_B T}{\hbar^2} \right)^{3/2} \int_0^\infty dx \frac{x^{1/2}}{z^{-1}e^x - 1} \equiv \frac{1}{\lambda_T^3} g_{3/2}(z) \quad (10)$$

where the function $g_{3/2}(z)$ is one of a class of functions that appear in the theory of Bose gases,

$$g_n(z) = \frac{1}{\Gamma(n)} \int_0^\infty dx \frac{x^{n-1}}{z^{-1}e^x - 1}. \quad (11)$$

As we decrease T in (10) keeping N/V fixed, z and hence $g_{3/2}(z)$ must both increase. But z can't take values greater than 1. When we reach $z = 1$, let's denote the temperature as $T = T_c$. We can determine T_c by setting $z = 1$ in equation (10)

$$T_c = \left(\frac{2\pi\hbar^2}{k_B m} \right) \left(\frac{N}{\zeta(3/2) V} \right)^{2/3}. \quad (12)$$

What happens if we decrease the temperature below T_c ? If we look at eq (10), it tells us that the particle number should decrease. But that's a stupid conclusion. It says the particle disappears when we make them cold.

We came to this wrong conclusion because if you look at equation (8) and (9) you can see the particles with 0 energy are not counted. we can use the Bose-Einstein distribution to compute the number of states in the ground state $E=0$,

$$n_0 = \frac{1}{z^{-1} - 1} \quad (13)$$

For most values of $z \in (0, 1)$, there are just a handful of particles sitting in the lowest state, and it doesn't matter if we miss them. But as z gets very very close to 1, then we get a macroscopic number of particles occupying the ground state.

If we redo our calculation, taking into account the particles in the ground state, then equation (9) is replaced by

$$N = \frac{V}{\lambda^3} g_{3/2}(z) + \underbrace{\frac{z}{1-z}}_{\text{\# particle in ground state}} \quad (14)$$

Now there's no problem keeping N fixed as we take z close to 1 because the additional term diverges. This means that if we have a finite N , then as we decrease T we can never reach $z = 1$. Instead, z must level out around $z \approx 1 - 1/N$ as $T \rightarrow 0$.

For $T < T_c$, the number of particles sitting in the ground state is $\mathcal{O}(N)$. Some simple algebra allows us to determine that the fraction of particles in the ground state is

$$\frac{n_0}{N} = 1 - \frac{V}{N \lambda^3} \frac{1}{\zeta(3/2)} = 1 - \left(\frac{T}{T_c} \right)^{3/2}. \quad (15)$$

At temperatures $T < T_c$, a macroscopic number of atoms discard their individual identities and merge into a collective entity — a single quantum state so large that it can be seen with the naked eye. This is known as the *Bose-Einstein condensate*.

3 Cooper Pair

To explain superconductivity Cooper considered bound states of two electron in momentum space. Cooper pair can be understood by considering two fermions on the top of a filled, non-interacting Fermi sphere at zero temperature. Then we can turn on an interaction between only these two electron. A contact interaction of strength U corresponds to hard balls with radius a_s scattering off each other. In quantum mechanical scattering theory, this corresponds to s -wave scattering, in which only wave functions with zero angular momentum ($l = 0$, hence the name s -wave) are considered to describe the scattered particles. This is appropriate for cold, dilute gases, where the typical scattering energies are low. Then, the contact interaction can be related to the scattering length (a_s) of s -wave scattering via

$$\frac{1}{U} = \frac{m}{4\pi\hbar^2 a_s} - \int \frac{d^3k}{(2\pi)^3} \frac{1}{2\epsilon_k} \quad (16)$$

where $\epsilon_k = \frac{\hbar^2 k^2}{2m}$ is the kinetic energy of the fermions. Derivation of this is given in appendix 7.2. It can be shown, that the binding energy is highest for pairs that have the highest possible number of final scattering states (density). The

binding energy is highest for a pair of opposite momentum fermions .
The binding energy for the composite pair of fermions with opposite momentum is given as

$$E_B = -\frac{8}{e^2} E_F e^{\pi/k_F |a_s|} \quad (17)$$

As you can see this is negative, compared to the Fermi energy, which is set to be 0.

I would like to draw one thing to your attention about the equation (16) Here we are dealing with attractive potential, so $1/U$ is negative. Also the 2nd term in the RHS is always positive. From this we can conclude two thing

- When $|U|$ is small $\implies \frac{1}{U}$ is large and -ve this makes a_s -ve
- When $|U|$ is large $\implies \frac{1}{U}$ is small and -ve this makes a_s +ve

Since the a_s is the scattering length. to compare this with the natural length scale of the system which is $\frac{1}{k_F} \propto$ interatomic distance, we will deal with term in the form $k_F a_s$. This allows us to separate the interaction in three different regimes which can be identified via the interaction parameter $1/(k_F a_s)$:

$$\underbrace{\text{BCS: } \frac{1}{k_F a_s} \ll -1,}_{\text{low attractive potential}} \quad \underbrace{\text{BEC: } \frac{1}{k_F a_s} \gg 1,}_{\text{large attractive potential}} \quad \text{BCS-BEC crossover: } -1 < \frac{1}{k_F a_s} < 1.$$

4 Ground state of the interacting fermi gas

The ground state of a BEC fermionic pairs is given as

$$|\Psi\rangle = \int \prod_i d^3 r_i \phi(\mathbf{r}_1 - \mathbf{r}_2) \Psi_{\uparrow}^{\dagger}(\mathbf{r}_1) \Psi_{\downarrow}^{\dagger}(\mathbf{r}_2) \cdots \phi(\mathbf{r}_{N-1} - \mathbf{r}_N) \Psi_{\uparrow}^{\dagger}(\mathbf{r}_{N-1}) \Psi_{\downarrow}^{\dagger}(\mathbf{r}_N) |0\rangle \quad (18)$$

with the fermion creation operator

$$\Psi_{\sigma}^{\dagger}(\mathbf{r}) = \sum_{\mathbf{k}} \frac{e^{-i\mathbf{k}\cdot\mathbf{r}}}{\sqrt{V}} c_{\mathbf{k}\sigma}^{\dagger}, \quad (19)$$

and the relative function

$$\phi(\mathbf{r}_1 - \mathbf{r}_2) = \sum_{\mathbf{k}} \frac{\phi_{\mathbf{k}}}{\sqrt{V}} e^{-i\mathbf{k}\cdot(\mathbf{r}_1 - \mathbf{r}_2)}. \quad (20)$$

Of course, the creation and annihilation operators of fermions obey the anti-commutation relations:

$$\{c_{\mathbf{k}\sigma}, c_{\mathbf{k}'\sigma'}\} = \{c_{\mathbf{k}\sigma}^{\dagger}, c_{\mathbf{k}'\sigma'}^{\dagger}\} = 0, \quad (21)$$

$$\{c_{\mathbf{k}\sigma}, c_{\mathbf{k}'\sigma'}^{\dagger}\} = \delta(\mathbf{k} - \mathbf{k}') \delta_{\sigma\sigma'}. \quad (22)$$

By defining the pair creation operator via

$$b^\dagger = \sum_{\mathbf{k}} \phi_{\mathbf{k}} c_{\mathbf{k}\uparrow}^\dagger c_{-\mathbf{k}\downarrow}^\dagger, \quad (23)$$

the state $|\Psi\rangle$ becomes formally identical to the Gross–Pitaevskii ground state of a condensate of bosons:

$$|\Psi\rangle = (b^\dagger)^{N/2} |0\rangle. \quad (24)$$

The commutation relations of the pair operator b are then given by

$$[b, b] = \sum_{\mathbf{k}\mathbf{k}'} \phi_{\mathbf{k}} \phi_{\mathbf{k}'} \left[c_{\mathbf{k}\uparrow}^\dagger c_{-\mathbf{k}\downarrow}^\dagger, c_{\mathbf{k}'\uparrow}^\dagger c_{-\mathbf{k}'\downarrow}^\dagger \right] = 0, \quad (25)$$

$$[b^\dagger, b^\dagger] = \sum_{\mathbf{k}\mathbf{k}'} \phi_{\mathbf{k}}^* \phi_{\mathbf{k}'}^* [c_{-\mathbf{k}\uparrow} c_{\mathbf{k}\downarrow}, c_{-\mathbf{k}'\uparrow} c_{\mathbf{k}'\downarrow}] = 0, \quad (26)$$

$$[b, b^\dagger] = \sum_{\mathbf{k}} |\phi_{\mathbf{k}}|^2 (1 - n_{\mathbf{k}\uparrow} - n_{\mathbf{k}\downarrow}). \quad (27)$$

This shows that b describes the creation and annihilation of bosons (composed of fermionic pairs). Later we will see (Fig 1b) that in the BEC limit the distribution function $n_{k\sigma}$ is vanishingly small and broad. In that case, the right-hand side of equation (27) becomes 1, as expected for the commutator relation of creation and annihilation operators for bosons.

It is more convenient to consider the grand canonical ensemble in which the particle number is not conserved but the chemical potential. Starting in the BEC limit, this corresponds to a coherent state of bosons described by the exponential of the same creation operator. Together with normalisation factors C and this yields:

$$|\Psi\rangle = C e^{\lambda b^\dagger} |0\rangle = C \prod_k \exp\left(\lambda \phi_k c_{k\uparrow}^\dagger c_{-k\downarrow}^\dagger\right) |0\rangle = C \prod_k \left(1 + \lambda \phi_k c_{k\uparrow}^\dagger c_{-k\downarrow}^\dagger\right) |0\rangle. \quad (28)$$

Choosing the normalisation factors to be $C = \prod_k u_k$ and $\lambda \phi_k = v_k/u_k$ together with the constraint $u_k^2 + v_k^2 = 1$ leads to the same form of the wave function as derived by Bardeen, Cooper and Schrieffer:

$$|\Psi_{\text{BCS}}\rangle = \prod_k \left(u_k + v_k c_{k\uparrow}^\dagger c_{-k\downarrow}^\dagger\right) |0\rangle. \quad (29)$$

This is remarkable: having started with the coherent state of bosons composed of fermionic pairs in the state of a bosonic condensate, one ends up with the BCS wave function! This implies that both regimes are described by the same wave function.

In order to fully determine the wave function, u_k and v_k have to be determined. This is done by minimizing the free energy of the system. Therefore, consider the Hamiltonian in the grand canonical ensemble with volume V and a contact interaction U :

$$H - \mu N = \sum_{k,\sigma} (\varepsilon_k - \mu) c_{k\sigma}^\dagger c_{k\sigma} + \frac{U}{V} \sum_{k,k'} c_{k\uparrow}^\dagger c_{-k\downarrow}^\dagger c_{k'\downarrow} c_{-k'\uparrow}, \quad \varepsilon_k = \frac{\hbar^2 k^2}{2m}. \quad (30)$$

The free energy F of the system is then given by

$$F = \langle \Psi | H - \mu N | \Psi \rangle = 2 \sum_k (\varepsilon_k - \mu) v_k^2 + \frac{U}{V} \sum_{k,k'} u_k v_k u_{k'} v_{k'}. \quad (31)$$

In order to minimize the free energy one has to take a derivative and set it to zero. The constraint on u_k and v_k allows the parametrization

$$u_k = \cos \theta_k, \quad v_k = \sin \theta_k.$$

Minimizing F using

$$\frac{\partial F}{\partial \theta_k} = 0$$

leads to

$$2(\varepsilon_k - \mu) u_k v_k + \frac{U}{V} (u_k^2 - v_k^2) \sum_{k'} u_{k'} v_{k'} = 0. \quad (32)$$

Now, defining the gap parameter Δ in the so-called gap equation

$$\Delta = \frac{U}{V} \sum_k \langle c_{k\uparrow} c_{-k\downarrow} \rangle = -\frac{U}{V} \sum_k u_k v_k, \quad (33)$$

so that equation (32) becomes

$$2(\varepsilon_k - \mu) u_k v_k - (u_k^2 - v_k^2) \Delta = 0. \quad (34)$$

The gap Δ is in fact the order parameter that characterises the transition from the Fermi gas to the superfluid phase, i.e. the BCS regime of the interacting Fermi gas. The solution for u_k and v_k of equation (34) is obtained by defining the quasiparticle energy

$$E_k = \sqrt{(\varepsilon_k - \mu)^2 + \Delta^2}.$$

Then,

$$u_k = \sqrt{\frac{1}{2} \left(1 + \frac{\varepsilon_k - \mu}{E_k} \right)}, \quad v_k = \sqrt{\frac{1}{2} \left(1 - \frac{\varepsilon_k - \mu}{E_k} \right)}. \quad (35)$$

Inserting these expressions for u_k and v_k into the gap equation gives

$$\Delta = -\frac{U}{V} \sum_k \frac{\Delta}{2E_k} \iff -\frac{1}{U} = \frac{1}{V} \sum_k \frac{1}{2E_k} \equiv \int \frac{d^3k}{(2\pi)^3} \frac{1}{2E_k} \quad (36)$$

where the last equality is understood as a continuous approximation of the sum by an integration over phase space. As mentioned above, the contact interaction U is related to the scattering length via equation ((16)). Replacing U via this relation then relates the gap to the scattering length:

$$\frac{1}{U} = \int \frac{d^3k}{(2\pi)^3} \frac{1}{2E_k} \iff \frac{m}{4\pi\hbar^2 a_s} = \int \frac{d^3k}{(2\pi)^3} \left(\frac{1}{2E_k} - \frac{1}{2\varepsilon_k} \right). \quad (37)$$

It should also be mentioned that an additional constraint on the system is given by the number equation:

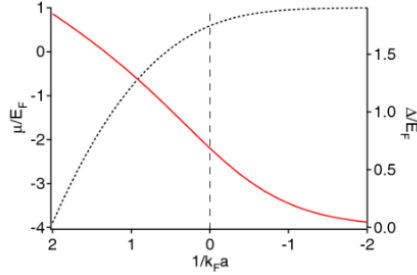
$$n = 2 \int \frac{d^3k}{(2\pi)^3} v_k^2. \quad (38)$$

Here one can identify $n_k = v_k^2$ as the occupation number density of momenta. In order to determine the gap and also the chemical potential of the system, it is useful to rewrite the gap and number equations in a dimensionless way using the following standard integrals:

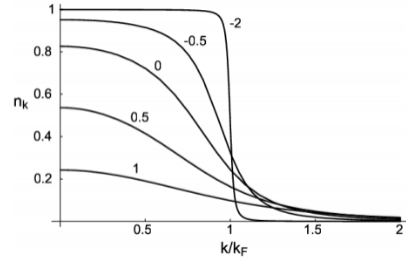
$$I_1 = \int_0^\infty dx \frac{1}{x^2} \left[\frac{1}{\sqrt{(x^2 - z)^2 + 1}} - 1 \right], \quad (39)$$

$$I_2 = \int_0^\infty dx x^2 \left[1 - \frac{x^2 - z}{\sqrt{(x^2 - z)^2 + 1}} \right]. \quad (40)$$

Using the Fermi energy $E_F = \frac{\hbar^2 k_F^2}{2m}$ and the Fermi momentum $k_F = (3\pi^2 n)^{1/3}$, the gap and number equation can then be written in the dimensionless form:



(a) Gap Δ (red) and chemical potential μ as a function of the scattering parameter. Positive values correspond to the BEC regime and negative to the BCS regime.



(b) Occupation number density $n_k = v_k^2$ for different interaction parameters.

$$\frac{1}{k_F a_s} = \frac{2}{\pi} \frac{\Delta}{E_F} I_1\left(\frac{\mu}{\Delta}\right), \quad (41)$$

$$1 = \frac{3}{2} \left(\frac{\Delta}{E_F} \right)^2 I_2\left(\frac{\mu}{\Delta}\right). \quad (42)$$

Further, inserting equation ((42)) into ((41)) yields the compact form:

$$\frac{1}{k_F a_s} = \frac{2}{\pi} \left[I_1\left(\frac{\mu}{\Delta}\right) - \frac{\mu}{3\Delta} I_2\left(\frac{\mu}{\Delta}\right) \right]. \quad (43)$$

The gap equation becomes

$$\frac{\Delta}{E_F} = \left[\frac{2}{3} \frac{\mu}{E_F} I_2\left(\frac{\mu}{\Delta}\right) \right]^{2/3}. \quad (44)$$

The gap Δ as a function of the scattering parameter $(k_F a_s)^{-1}$ is now obtained by solving the gap equation ((43)) for μ/Δ and inserting this expression into the number equation ((44)). Figure 1a shows the result for the gap and the chemical potential as a function of the scattering parameter. We can see $\mu < 0$ in the BEC regime as expected.

Three regimes can be identified via the interaction parameter $1/(k_F a_s)$:

$$\text{BCS: } \frac{1}{k_F a_s} \ll -1, \quad \text{BEC: } \frac{1}{k_F a_s} \gg 1, \quad \text{BCS-BEC crossover: } -1 < \frac{1}{k_F a_s} < 1.$$

Also, having obtained Δ , it can be inserted into $n_k = v_k^2$. Figure 1b shows n_k for different scattering parameters, i.e., different regimes of the Fermi gas. In the BCS regime it closely resembles a step function, just like the ideal Fermi gas, and it becomes broader and flatter in the BCS-BEC crossover region.

4.1 Low Energy Excitation

To study the low energy excitation we can change the physical picture still keeping the physics unchanged. We consider the Bogoliubov transformation of creation and annihilation operators which leaves the anti-commutator relations unchanged:

$$\begin{pmatrix} c_{k\uparrow} \\ c_{-k\downarrow}^\dagger \end{pmatrix} = \begin{pmatrix} u_k & -v_k \\ v_k & u_k \end{pmatrix} \begin{pmatrix} \gamma_{k\uparrow} \\ \gamma_{-k\downarrow}^\dagger \end{pmatrix} \quad (\text{Bogoliubov transformation}) \quad (45)$$

$$\{\gamma_{k\sigma}, \gamma_{k'\sigma'}^\dagger\} = \{c_{k\sigma}, c_{k'\sigma'}^\dagger\} = \delta(k - k') \delta_{\sigma\sigma'}, \quad (46)$$

$$\{\gamma_{k\sigma}, \gamma_{k'\sigma'}\} = \{\gamma_{k\sigma}^\dagger, \gamma_{k'\sigma'}^\dagger\} = 0. \quad (47)$$

It is non-trivial to show that these u_k and v_k are actually the same as in the previous derivation of the gap equation. But indeed, the Bogoliubov transformation can be understood as a rotation in a plane and in this case as a rotation in a plane spanned by the c creation and annihilation operators. The γ operators are then linear combinations of the c operators describing fictive or quasi-particles. This also justifies the earlier approach to parametrize the u_k and v_k by $\cos(\theta_k)$ and $\sin(\theta_k)$, respectively. Applying the transformation to the Hamiltonian gives:

$$H - \mu N = -\frac{V}{U} \sum_k \Delta^2 + \sum_k (\epsilon_k - \mu - E_k) + \sum_k E_k \left(\gamma_{k\uparrow}^\dagger \gamma_{k\uparrow} + \gamma_{k\downarrow}^\dagger \gamma_{k\downarrow} \right). \quad (48)$$

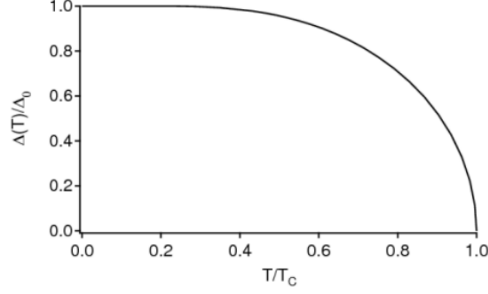


Figure 2: Shown is the gap in units of the gap at zero temperature as a function of temperature in units of the critical temperature, below which condensation of fermionic pairs into a BEC occur.

At finite temperature the pairing field becomes:

$$\langle c_{k\uparrow} c_{-k\downarrow} \rangle = -u_k v_k \left[1 - \langle \gamma_{k\uparrow}^\dagger \gamma_{k\uparrow} \rangle - \langle \gamma_{k\downarrow}^\dagger \gamma_{k\downarrow} \rangle \right]. \quad (49)$$

Since the quasi-particles follow Fermi-Dirac statistics $\langle \gamma_{k\uparrow}^\dagger \gamma_{k\uparrow} \rangle = \frac{1}{1+e^{\beta E_k}}$, the gap equation $\Delta = \frac{U}{V} \sum_k \langle c_{k\uparrow} c_{-k\downarrow} \rangle$ becomes temperature dependent (with $\beta = 1/k_B T$).

$$\begin{aligned} \Delta &= -\frac{U}{V} \sum_k u_k v_k \left(1 - \frac{2}{1+e^{\beta E_k}} \right) \\ &= -\frac{U}{V} \sum_k \frac{1}{2} \left(\frac{\epsilon_k - \mu}{E_k} \right) \left(1 - \frac{2}{1+e^{\beta E_k}} \right) \\ &= -\frac{U}{V} \sum_k \frac{1}{2} \left(1 - \frac{\epsilon_k - \mu}{E_k} \right) \left(1 - \frac{2}{1+e^{\beta E_k}} \right) \\ &\Rightarrow 1 = -\frac{U}{V} \sum_k \left(\frac{1}{2E_k} \right) \left(1 - \frac{2}{1+e^{\beta E_k}} \right) \\ &\Rightarrow -\frac{1}{U} = \int \frac{d^3 k}{(2\pi)^3} \left(\frac{1}{2E_k} \right) \left(1 - \frac{2}{1+e^{\beta E_k}} \right) \\ &= \int \frac{d^3 k}{(2\pi)^3} \left(\frac{1}{2E_k} \right) \tanh \left(\frac{\beta E_k}{2} \right). \end{aligned}$$

Now using eq (16) we get

$$-\frac{m}{4\pi \hbar a_s} + \int \frac{d^3 k}{(2\pi)^3} \frac{1}{2\epsilon_k} = \int \frac{d^3 k}{(2\pi)^3} \frac{1}{2E_k} \tanh(\beta E_k)$$

The gap equation is given by

$$-\frac{m}{4\pi\hbar a_s} = \int \frac{d^3k}{(2\pi)^3} \left(\frac{1}{2E_k} \tanh(\beta E_k/2) - \frac{1}{2\epsilon_k} \right) \quad (50)$$

Solving the temperature-dependent gap equation yields

$$\Delta(T) \approx \begin{cases} \Delta_0 - \sqrt{2\pi\Delta_0 k_B T} \exp\left(-\frac{\Delta_0}{k_B T}\right) & \text{for } T \ll T_C \\ \sqrt{\frac{8\pi^2}{7\zeta(3)} k_B T_C} \sqrt{1 - \frac{T}{T_C}} & \text{for } T_C - T \gg T_C \end{cases} \quad (51)$$

where

$$\Delta_0 = \frac{8}{e^2\pi} E_F \exp\left(-\frac{\pi}{2k_F|a_s|}\right) \quad (52)$$

is the zero-temperature gap. Figure 2 shows the temperature-dependent gap below the critical temperature, below which condensation takes place.

4.2 Pair size

Now we will look at the pair size of these bound fermions.[4]. The size of the pairs may be estimated using

$$\xi_{\text{pair}}^2 = \frac{\langle \phi_k | r^2 | \phi_k \rangle}{\langle \phi_k | \phi_k \rangle} \quad (53)$$

The real-space pair correlation function is defined as

$$\psi(r) = \langle 0 | c_\downarrow(r) c_\uparrow(0) | \Psi_{\text{BCS}} \rangle, \quad | \Psi_{\text{BCS}} \rangle = \prod_k \left(u_k + v_k c_{k\uparrow}^\dagger c_{-k\downarrow}^\dagger \right) | 0 \rangle. \quad (54)$$

This quantity gives the amplitude to find one electron with spin $\downarrow$ at r and one with spin $\uparrow$ at $r = 0$. We can write $c_\sigma(r)$ in terms of $c_{k\sigma}$

$$c_\sigma(r) = \frac{1}{\sqrt{V}} \sum_k e^{i\mathbf{k}\cdot\mathbf{r}} c_{k\sigma}$$

Putting this into equation (54) and using the relation $\langle 0 | c_{-k\downarrow} c_{k\uparrow} | \Psi_{\text{BCS}} \rangle = u_k v_k = \frac{\Delta}{2E_k}$

$$\Psi(\mathbf{r}) = \frac{1}{V} \sum_{\mathbf{k}} e^{i\mathbf{k}\cdot\mathbf{r}} \frac{\Delta}{2E_k},$$

From this equation we get the ϕ_k used in eq (53) we get

$$|\phi_{\mathbf{k}}\rangle = u_k v_k e^{i\mathbf{k}\cdot\mathbf{r}} = \frac{\Delta}{2E_k} e^{i\mathbf{k}\cdot\mathbf{r}},$$

so $\phi_k = \frac{\Delta}{2E_k}$, which is the amplitude of finding cooper pair with momemntum k and $-k$. Converting the $r^2 \rightarrow \nabla_k^2$ (writing the position oerator in momemntum space) eq (53) can be written as

$$\xi_{\text{pair}}^2 = \frac{\int d^3k |\nabla_{\mathbf{k}} \phi_{\mathbf{k}}|^2}{\int d^3k |\phi_{\mathbf{k}}|^2}$$

In the BEC limit: $\mu < 0$ (large and negative) see figure 1a

$$E_k = \sqrt{(\epsilon_k + |\mu|)^2 + \Delta^2}$$

$$\phi_k = \frac{\Delta}{2(\epsilon_k + |\mu|)}$$

Evaluating the denominator

$$\int d^3k \left| \frac{\Delta}{2(\epsilon_k + |\mu|)} \right|^2 = \frac{\Delta^2}{4} \int d^3k \frac{1}{(\epsilon_k + |\mu|)^2}, \epsilon_k = \frac{\hbar^2 k^2}{2m},$$

$$\int_0^\infty d\epsilon_k \sqrt{\epsilon_k} \frac{1}{(\epsilon_k + |\mu|)^2} \propto \frac{1}{\sqrt{|\mu|}} \propto a_s$$

Now calculating the numerator

$$\int_0^\infty d\epsilon_k \epsilon_k \sqrt{\epsilon_k} \frac{1}{(\epsilon_k + |\mu|)^4} \propto \frac{1}{|\mu|^{3/2}} \propto a_s^3$$

Putting everything together we get

$$\xi_{\text{pair}}^2 \propto a_s^2,$$

$$\xi_{\text{pair}} \sim a_s \quad (\text{BEC limit}),$$

$$\boxed{\xi_{\text{pair}} \sim a_s \ll k_F^{-1} \quad (\text{pair size} \ll \text{interparticle spacing}).}$$

In the BCS regime ,

$$\text{denominator: } \int d^3k |\phi_k|^2 \propto \int_0^\infty d\epsilon_k \frac{\Delta^2}{4(\epsilon_k^2 + \Delta^2)}$$

$$= \frac{\Delta}{4} \left[\tan^{-1} \left(\frac{\epsilon_k}{\Delta} \right) \right]_0^\infty = \frac{\pi \Delta}{8}$$

For $\frac{\Delta}{E_F} \ll 1$ see figure 1a, $|a_s| k_F \rightarrow \infty$ (strong, weakly attractive)

$$v_F = \frac{1}{\hbar} \left. \frac{\partial \epsilon_k}{\partial k} \right|_{k=k_F} = \frac{\hbar k_F}{m}$$

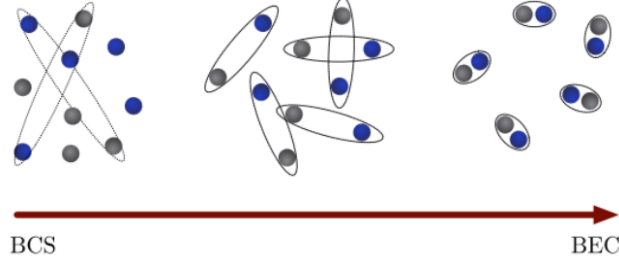


Figure 3: This figure illustrates the size of fermionic pairs in different regimes. In the BEC limit, the size of pairs is much smaller than the typical distance between pairs. In the BCS limit, pairs are strongly overlapping, which leads to properties of a superfluid rather than a condensate.

$$\begin{aligned}\nabla_k \phi_k &= \frac{d\epsilon_k}{dk} \frac{d\phi_k}{d\epsilon_k} = \frac{\hbar^2 k}{m} \frac{d}{d\epsilon_k} \left(\frac{\Delta}{2\sqrt{\epsilon_k^2 + \Delta^2}} \right) \\ &\Rightarrow (\hbar\nu_k) \frac{\Delta\epsilon_k}{2(\epsilon_k^2 + \Delta^2)^{3/2}} \\ \text{Numenator: } \int d^3k |\nabla_k \phi_k|^2 &\propto (\hbar\nu_F)^2 \int d\epsilon_k \frac{\Delta^2 \epsilon_k^2}{(\epsilon_k^2 + \Delta^2)^3} \sim \frac{(\hbar\nu_F)^2}{4\Delta}\end{aligned}$$

Putting everything together we have

$$\xi_{pair}^2 \sim \left(\frac{\hbar\nu_F}{\Delta} \right)^2 \Rightarrow \xi_{pair} = \frac{\hbar\nu_F}{\Delta} = \frac{E_F}{\Delta k_F} \quad (55)$$

Since in the BCS regime $\frac{\Delta}{E_F} \ll 1$. we have

$$\xi_{pair} \gg \frac{1}{k_F} \propto \text{interatomic distance}$$

Size of the pair much larger than interparticle distance

figure 3 gives an intuitive picture of the regimes of the interacting Fermi gas defined by the interaction parameter.

5 Fermionic Pair Formation and Condensation

5.1 Pair Formation Temperature

The gap is the order parameter of the Fermi gas characterizing the phase transition from the phase of unpaired fermions to the phase of paired fermions. The temperature T_{pair} at which fermionic pairs start to form is determined by

the condition $\Delta = 0$, which when inserted into the temperature dependent gap equation(50) yields an equation for the pairing temperature:

$$\frac{-m}{4\pi\hbar^2 a_s} = \int \frac{d^3k}{(2\pi)^3} \left(\frac{1}{2(\epsilon_k - \mu)} \tanh\left(\frac{\beta_{pair}(\epsilon_k - \mu)}{2}\right) - \frac{1}{2\epsilon_k} \right) \quad (56)$$

Since the system can be expected to follow the Fermi-Dirac statistic above T_{pair} , the number equation is

$$n = 2 \int \frac{d^3k}{(2\pi)^3} \frac{1}{1 + \exp(\beta_{pair}(\epsilon_k - \mu))}$$

solving these two equations we get T_{pair} . In the BCS regime it can be assumed that $\mu \gg k_B T_{pair}$, hence $\mu \approx E_F$. Solving the above equation with this chemical potential leads to

$$T_{pair}^{BCS} = T_C^{BCS} = \frac{8e^\gamma}{\pi e^2} \exp\left(-\frac{\pi}{2k_F |a_s|}\right) \quad (57)$$

In the BCS regime, the pair formation and critical temperature are the same, because in that regime the Fermi fluid constitutes a fluid of strongly overlapping fermion pairs, whose inter-particle spacing is much larger than the average particle distance and which do not form a condensate in the sense of bosons in a BEC. The overlapping pairs rather exhibit a behaviour similar to superconductivity. In fact, superconductivity can be seen in a more general picture as so-called charged superfluidity and the interacting Fermi gas below the critical temperature forms a superfluid, i.e, a fluid without any viscosity.

In the BEC regime, T_{pair}^{BCS} is given by

$$T_{pair}^{BCS} = \frac{1}{2} \frac{|E_B|}{W[(\frac{\pi}{6})^{\frac{1}{3}} \frac{E_B}{2E_F}]} \quad (58)$$

where $W(x) \approx \ln(x - \ln[\ln(x)])$. This leads to a very steep increase of T_{pair} towards the BEC regime. Overall, T_{pair} increases exponentially from zero in the BCS regime ($1/k_F a_s \rightarrow -\infty$) to very large values in the BEC limit, i.e, in the limit of $1/k_F a_s \rightarrow \infty$. This can intuitively be understood as follows: deep in the BEC regime, the very large positive scattering length leads to the formation of very strongly bound fermionic pairs. Hence, the larger the scattering length, the larger has to be the energy, ie, temperature, to break up these pairs, which defines the temperature of pair formation.

5.2 Critical Temperature for Condensation

Deep in the BEC regime the condensation temperature is given by the critical temperature discussed earlier eq(12) but now for bosons with molecular density $n_M = n/2$ and mass $m_M = 2m$

$$T_C^{BEC} = \frac{2\pi\hbar^2}{n_M} \left(\frac{n_M}{\zeta(3/2)} \right)^{2/3} = 0.22 E_F \quad (59)$$

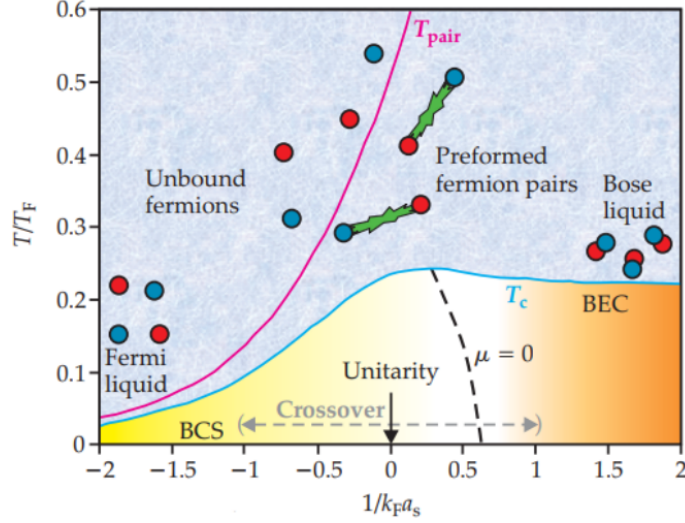


Figure 4: The figure shows T_{pair} and T_C as functions of the interaction parameter.

However there are small correction to this value . To get the exact condensation curve as shown in the fig 4 we have to solve this in a bit different way and that's what we do in the next section

6 Pair Formation and Condensation through functional integration formulation

Let us consider a continuum model where the Hamiltonian density is given as

$$\mathcal{H} = \bar{\psi}_\sigma(x) \left[-\frac{\nabla^2}{2m} - \mu \right] \psi_\sigma(x) - g \bar{\psi}_\uparrow(x) \bar{\psi}_\downarrow(x) \psi_\downarrow(x) \psi_\uparrow(x)$$

we choose units in which $\hbar = k_B = 1$ and the system is in a unit volume. Here ψ and $\bar{\psi}$ are c-numbers which are replaced by the field operators. The partition function is given by $Z = \int D[\bar{\phi}, \phi] \exp(-S)$ where the action is given as

$$S = \int_0^\beta d\tau \int dx (\bar{\psi}_\sigma \partial_\tau \psi_\sigma(x) + \mathcal{H}) \quad (60)$$

Decoupling the fermion interaction with the Hubbard-Stratonovich field $\Delta(x, \tau)$ which couples to $\bar{\psi}\bar{\psi}$ and integrating out the fermions we obtain $Z = \int D\Delta D\bar{\Delta} \exp(-S_{eff}[\Delta, \bar{\Delta}])$. The effective action is given as

$$S_{eff}[\Delta(x)] = \frac{1}{g} \int_0^\beta d\tau \int dx |\Delta(x)|^2 - \text{Tr} \ln G^{-1}[\Delta(x)] \quad (61)$$

where $G^{-1}(x)$ is the inverse Nambu propagator

$$\mathbf{G}^{-1}(x, x') = \begin{pmatrix} -\partial_\tau + \nabla^2/2m + \mu & \Delta(x) \\ \Delta^*(x) & -\partial_\tau - \nabla^2/2m - \mu \end{pmatrix} \delta(x - x'). \quad (62)$$

We now analyze this functional integral in terms of static, uniform saddle-points and fluctuations about these. The trivial saddle-point $\Delta = 0$, which is stable at sufficiently high T for all couplings, will become unstable below a certain temperature, which we denote, for now, by T_0 . The transition temperature is then defined by the solution of $\delta S_{eff}/\delta \Delta[\Delta = 0] = 0$ which can be written as

$$\frac{1}{g} = \sum_k \tanh\left(\frac{E_k}{2T_0}\right)/2E_k \quad (63)$$

where $E_k = \epsilon_k - \mu$. In the above equation we have ultraviolet divergence. The idea is to replace the bare g by the low energy limit of the two-body T-matrix. In three dimension (3D) we use

$$\frac{m}{4\pi a_s} = -\frac{1}{g} + \sum_{|k| < \Lambda} (2\epsilon_k)^{-1} \quad (64)$$

This equation is very similar to eq(16). Using (64) we obtain the equation for the transition temperature in terms of the renormalized coupling a_s :

$$-\frac{m}{4\pi a_s} = \sum_k \left[\frac{\tanh(E_k/2T_0)}{2E_k} - \frac{1}{2\epsilon_k} \right] \quad (65)$$

where $\epsilon_k = |k|^2/2m$. There are two unknowns in the equation, T_0 and μ and thus we need another equation $N = -\frac{\partial \Omega}{\partial \mu}$ which fixes the chemical potential for a given density.

6.1 Saddle Point Approximation

The saddle-point estimate T_0 is obtained by using the approximation $\Omega_0 = S_{eff}[\Delta = 0]/\beta$ for the thermodynamic potential. The number equation is then given by

$$n = n_0(\mu, T) \equiv \sum_{\mathbf{k}} \left(1 - \tanh\left(\frac{\xi_{\mathbf{k}}}{2T}\right) \right). \quad (66)$$

In the weak coupling limit, $1/k_F a_s \rightarrow -\infty$ and we find the BCS results $\mu = \epsilon_F$ and

$$T_0 = 8e^{-2}\gamma\pi^{-1}\epsilon_F \exp\left(-\frac{\pi}{2k_F|a_s|}\right),$$

where $\gamma \simeq 1.781$.

In the strong coupling limit i.e In this limit $1/k_F a_s \rightarrow +\infty$ and one finds tightly bound pairs with binding energy $E_b = 1/ma_s^2$. The non-degenerate Fermi system has $\mu \simeq -E_b/2$ and one finds

$$T_0 \simeq \frac{E_b}{2 \ln(E_b/\epsilon_F)^{3/2}}.$$

We have found similar results previously. But the saddle point approximation becomes progressively worse with increasing coupling. Though this gives us the pair formation temperature but to get the condensation temperature, we have to consider quantum fluctuation on the top of saddle point approximation.

6.2 Beyond Saddle point

Since the saddle-point estimate for condensation temperature (T_c) with increasing coupling, we must include the effects of fluctuations about $\Delta(x, \tau) = 0$. After including the dynamic fluctuation term which retains the full frequency dependency. In crude sense, the formation of tightly bound pairs is a quantum fluctuation which must be included in the functional integral. The effective action is now given as

$$S_{eff} \approx S_{eff}[0] + \sum_{q, iq_l} |\Delta(q, iq_l)|^2 / \Gamma(q, iq_l) \quad (67)$$

where

$$\Gamma^{-1}(q, iq_l) = \sum_k \left[\frac{1 - n_k - n_{k+q}}{iq_l - E_k - E_{k+q}} \right] - \frac{m}{4\pi a_s} \quad (68)$$

Here n_k is a Fermi function and $iq_l = i2l\pi/\beta$ are the Matsubara bose frequencies. The thermodynamic potential is $\Omega = \Omega_0 - \beta^{-1} \sum_{q, iq_l} \ln \Gamma(q, iq_l)$. The new number equation $N = -\frac{\partial \Omega}{\partial \mu}$ is given as

$$n = n_0(\mu, T) + \sum_q \int_{-\infty}^{+\infty} \frac{d\omega}{\pi} n_B(\omega) \frac{\partial \delta}{\partial \mu}(q, \omega) \quad (69)$$

where $n_B(\omega) = \frac{1}{\exp(\beta\omega) - 1}$ is the bose function.

The temperature T_c at which long range order is established is defined by the solution of eq(65) and eq(69). In weak coupling, the results are essentially unaffected by the inclusion of Gaussian fluctuations in the number equation and T_c is the same as T_0 obtained earlier. with increasing interaction parameter, T_c deviates from T_0 and goes to a finite strong coupling limit which coincides with the BEC transition temperature. (see Figure 4)

7 Appendix

7.1 Relation of the scattering length with scattering amplitude

The scattering length is defined as the scattering amplitude at low energy. The incoming particle with low energy (high wavelength) won't see the structure and hence scatter off uniformly in all direction. ie, mathematically we are considering only the S-wave scattering, where the scattering amplitude is not dependent on the direction. In scattering theory the asymptotic wavefunction is given as (we

assume there is an finite ranged scatterer at the origin and there is an incoming plane wave along the z-axis):

$$\psi(r, \theta) = e^{ikz} + f(\theta) \frac{e^{ikr}}{r} \quad (70)$$

We can expand the plane wave in terms of spherical waves and Legendre polynomials $P_l(\cos\theta)$:

$$e^{ikz} \approx \frac{1}{2ikr} \sum_{l=0}^{\infty} (2l+1) P_l(\cos\theta) [(-1)^{l+1} e^{-ikr} + e^{ikr}] \quad (71)$$

for s-wave we set $l=0$, we get

$$e^{ikz} = \frac{1}{2ikr} (e^{ikr} - e^{-ikr}) \quad (72)$$

now for the s-wave $\psi(r)$ can be written as

$$\Psi(r) = \frac{1}{2ikr} (e^{ikr} - e^{-ikr}) + f \frac{e^{ikr}}{r} \quad (73)$$

The s-wave radial solution is given as

$$\Psi_s(r) = \frac{A}{r} \sin(kr + \delta_s) = \frac{A}{2ir} (e^{i\delta_s} e^{ikr} - e^{-i\delta_s} e^{-ikr}) \quad (74)$$

Matching the incoming spherical wave coefficient , we get

$$\frac{-1}{2kr} = \frac{-A}{2ir} e^{-i\delta_s} \implies A = \frac{e^{i\delta_s}}{k} \quad (75)$$

$$\Psi_s(r) = \frac{e^{i\delta_s}}{2ik} (e^{iks-i\delta_s} - e^{-iks-i\delta_s}) \quad (76)$$

$$\Psi_s(r) = \frac{1}{2ik} (e^{2i\delta_s} e^{iks} - e^{-iks}) \quad (77)$$

Comparing with the (73) we identify the scattering amplitude as :

$$f = \frac{e^{2i\delta_s} - 1}{2ik}. \quad (78)$$

at low energy we can approximate it as

$$f = \frac{\delta_s}{k} = -a_s \quad (79)$$

7.2 Relation between U (scattering potential) and a_s (scattering length)

Let us consider the scattering of two electron in the center of mass frame so that both of the electron have the same energy and momentum. The T matrix in scattering is given by

$$T_{ni} = V_{ni} + \frac{1}{\hbar} \sum_m \frac{V_{nm} T_{mi}}{E_i - E_m + i\hbar\epsilon} \quad (80)$$

$$T_{k'k}(E) = V_{k'k} + \frac{1}{\hbar} \sum_q \frac{V_{k'q} T_{qk}}{E - E_q + i\epsilon} \quad (81)$$

Here ϵ is a very small number so we have set that to 0. If we work in the COM frame then both the incoming electron have the same momentum and hence the same energy, so replacing $E_q = 2\epsilon_q$ with each electron carrying energy ϵ_q . Here, E is the total energy of the incoming electrons. Then the above equation in the continuum limit becomes

$$T_{k',k}(E) = V + \frac{1}{\hbar} \left(\int \frac{d^3q}{(2\pi)^3} \frac{1}{E - 2\epsilon_q} \right) V T(E) \quad (82)$$

Here, we have assumed that the potential V does not have any momentum dependency. As a result, the T matrix does not have any momentum dependency, which is completely logical if we consider only s-wave scattering, where there is no θ or ϕ dependency

$$T(E) = V + V \Pi(E) T(E) \quad (83)$$

where $\Pi(E) = \frac{1}{\hbar} \left(\int \frac{d^3q}{(2\pi)^3} \frac{1}{E - 2\epsilon_q} \right)$

$$T(E) = \frac{V}{1 - V \Pi(E)} \quad (84)$$

The scattering amplitude is given as

$$f(k', k) = -\frac{mL^3}{2\pi\hbar^2} \langle k' | U | \psi^{(+)} \rangle = -\frac{mL^3}{2\pi\hbar^2} \langle k' | T | k \rangle \quad (85)$$

where $\psi^{(+)} = |i\rangle + \frac{1}{E_i - H_0 + i\hbar\epsilon} V \psi^{(+)}$ is the Lippmann-Schwinger equation

$$f(E) = -\frac{mL^3}{2\pi\hbar^2} T(E) \quad (86)$$

At low temperature we get

$$f(0) = -\frac{mL^3}{2\pi\hbar^2} T(0) \quad (87)$$

defining the low temperature scattering amplitude, in s-wave scattering, as the scattering length a_s , we have

$$a_s = -\frac{mL^3}{2\pi\hbar^2} T(0) \quad (88)$$

$$T(0) = \frac{4\pi\hbar^2 a_s}{m} \quad (89)$$

Putting this in equation (84)

$$\frac{V}{1 - V\Pi(0)} = \frac{4\pi\hbar^2 a_s}{m} \quad (90)$$

$$\frac{1}{V} = \Pi(0) + \frac{m}{4\pi\hbar^2 a_s} \quad (91)$$

so finally we have

$$\boxed{\frac{1}{V} = \frac{m}{4\pi\hbar^2 a_s} - \int \frac{d^3q}{(2\pi)^3} \frac{1}{2\epsilon_q}} \quad (92)$$

References

- [1] Jörg Holsten. Fermionic pairs and the bcs-bec crossover. Summary for the Statistical Physics Seminar, Heidelberg University, Summer Term 2020. Available online at https://www.thphys.uni-heidelberg.de/~wolschin/statsem20_11s.pdf.
- [2] Wolfgang Ketterle. Fermi gases, bec-bcs crossover. YouTube video, MIT 8.422 Atomic and Optical Physics II, Spring 2013, July 2014. Accessed on November 23, 2025, https://www.youtube.com/watch?v=0_zjGYvP4Ps&t=3268s.
- [3] Mohit Randeria, Ji-Min Duan, and Lih-Yir Shieh. Superconductivity in a two-dimensional fermi gas: Evolution from cooper pairing to bose condensation. *Phys. Rev. B*, 41:327–343, Jan 1990.
- [4] Mohit Randeria and Edward Taylor. Crossover from bardeen-cooper-schrieffer to bose-einstein condensation and the unitary fermi gas. *Annual Review of Condensed Matter Physics*, 5(1):209–232, March 2014.
- [5] David Tong. Quantum gases. Lecture notes, Statistical Physics, University of Cambridge. Available online at <https://www.damtp.cam.ac.uk/user/tong/statphys/three.pdf>.